

Exercise Sheet 1

Key:

- **Questions marked with a ☕** are for discussion in your tutorials. Please complete these questions before your tutorial.
- **Questions marked with a ✨** are difficult questions.
- **Questions marked with a 🎵** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

🎵 **Exercise 1.** Let $m \neq 0$ be a real number.

- (i) Prove that the line

$$y = mx + \frac{15}{4m}$$

is tangent to the parabola $y^2 = 15x$, for all non-zero values of m .

- (ii) Using this result, or otherwise, find the equations of the common tangents to this parabola and the circle $x^2 + y^2 = 16$.

☕ **Exercise 2.** A circle is given by the equation

$$x^2 - 6x + y^2 - 8y = 0.$$

- (i) Find the radius and the coordinates of the centre of the circle.
- (ii) Find the values of c for which the line

$$y = \frac{3}{4}x + c$$

is tangent to the circle.

☕ **Exercise 3.** Consider the sets

$$A = \left\{ (x, y) \in \mathbb{R}^2 : \frac{x^2}{4} + \frac{y^2}{9} \leq 1 \right\},$$

$$B = \{(x, y) \in \mathbb{R}^2 : y \geq x^2\}$$

and

$$C = \{(x, y) \in \mathbb{R}^2 : y \leq x\}.$$

Sketch/describe the regions A , B and C and hence sketch $A \cap B$ and $A \cap C$.

☛ **Exercise 4.** Consider the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

with parameterization

$$(x, y) = (a \sec \theta, b \tan \theta), \quad \theta \in [-\pi, \pi], \quad \theta \neq \pm \frac{\pi}{2}.$$

Show that the equations of the tangent and normal (at parameter location θ) are given by

$$bx \sec \theta - ay \tan \theta = ab$$

and

$$by \sec \theta + ax \tan \theta = (a^2 + b^2) \sec \theta \tan \theta.$$

♪ **Exercise 5.** Sketch the ellipse

$$\frac{x^2}{2} + 2y^2 = 1$$

and find the equation of the tangent and normal through the point $P : (x, y) = (1, 1/2)$. Show these two lines on your sketch.

♪ **Exercise 6.** Consider the parabola $y^2 = 4(x - 1)$.

- (i) Find the values of m for which the line $y = mx$ is tangent to the parabola.
- (ii) For these values of m , state the coordinates at which the tangent meets the parabola.
- (iii) Consider the line $y = m(x - 2)$ and suppose this meets the parabola.

Find the quadratic equation satisfied by the x -coordinate(s) of intersection.

Hence by considering the number of roots of this equation, investigate whether the line $y = m(x - 2)$ can ever be tangent to the parabola.

♪ **Exercise 7.** Sketch the following curves.

- (i) $r = \frac{1}{1 - \sin \theta}$
- (ii) $r = 1 + \cos \theta$
- (iii) $r = \theta$
- (iv) $r = \sin \theta$
- (v) $r = \sin(2\theta)$

☛ **Exercise 8.** Prove that the shortest distance between the point $P : (x_0, y_0)$ and the point N on the line

$$L : ax + by + c = 0$$

is given by

$$|PN| = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}.$$

✚ **Exercise 9.** The curve defined by

$$r = \frac{l}{1 + e \cos \theta},$$

is a conic section. Here, l is a positive constant and $e \geq 0$ is the eccentricity.

- (i) Show that if $e = 1$, the equation of the curve can be written in the form $x = py^2 + q$ where p and q are constants.
- (ii) Show that if $e < 1$ the curve is an ellipse.

Remark: if $e > 1$, the curve is a hyperbola. If one discards values of θ leading to negative r the curve is one branch of a hyperbola. The curve includes both branches if one identifies $(-r, \theta)$ with $(r, \theta - \pi)$ (this convention is found in some textbooks and employed by several graphical plotters).

✚ **Exercise 10.** Plot the curve given parametrically by

$$x = \cos t, \quad y = \frac{|\sin t|}{t} \quad \text{for } -2\pi \leq t \leq 2\pi.$$

Hint: consider carefully what happens as t passes through zero.